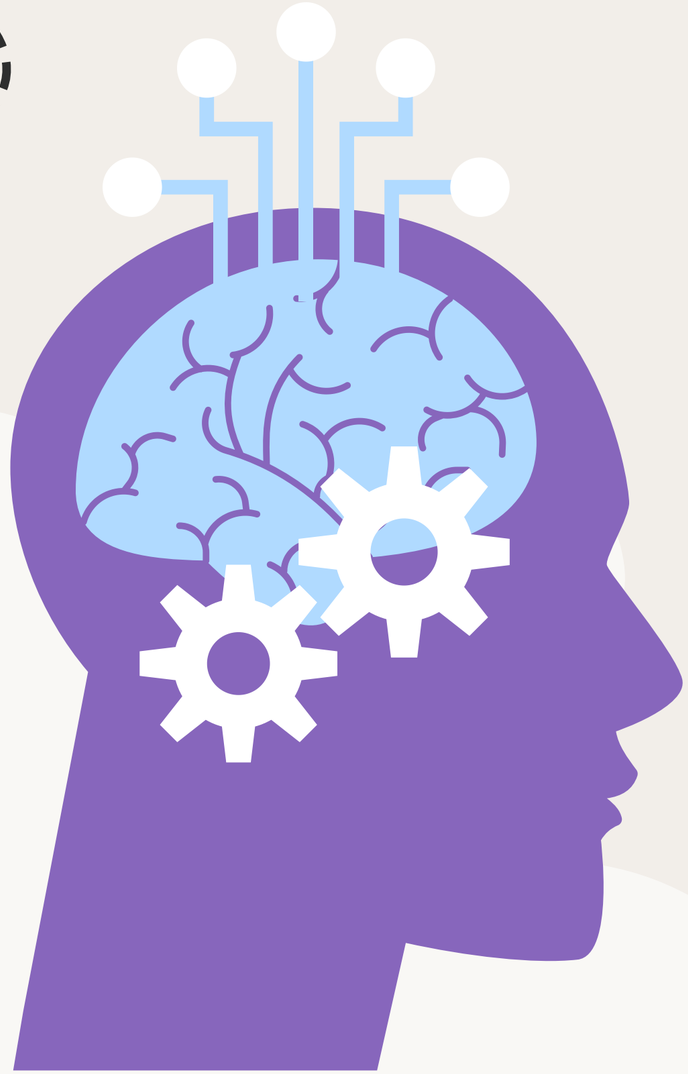

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Vision-LLM

Beyond Binary Emotions

A Multi-Architecture Vision-Language Approach for Compound Facial Expression Recognition

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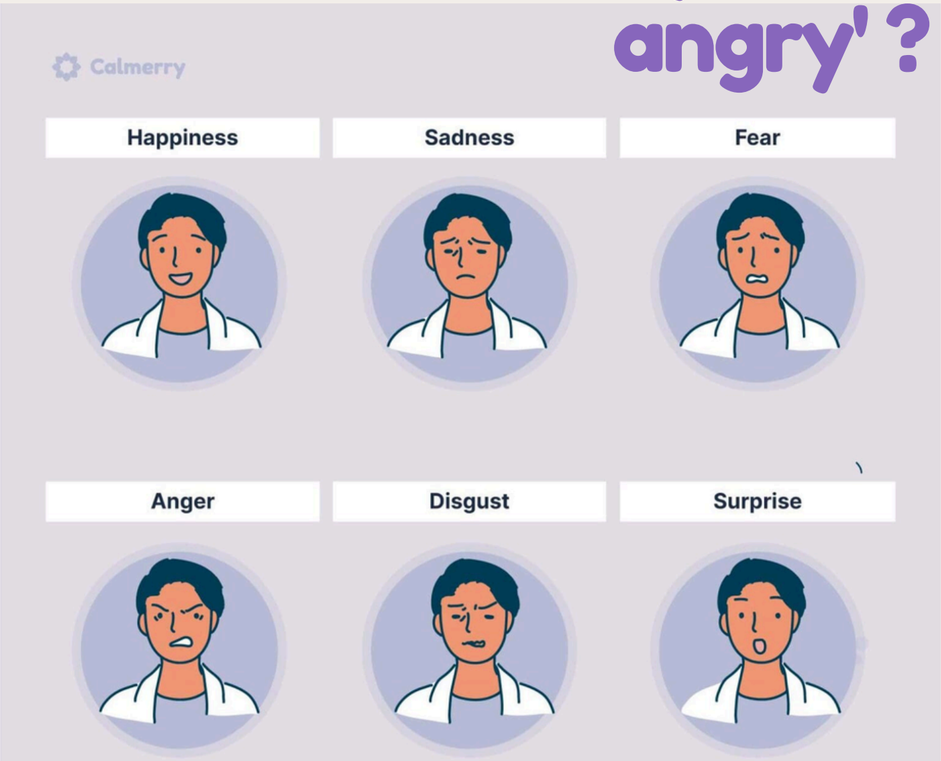
Why Compound Emotions?

Can you tell the difference between 'sadly fearful' and 'sadly angry' ?



Ekman's 6 basic emotions

Real life is more complex

Compound emotions

> *Applications :*
>
> Social Robotics
>
> Clinical Psychology
>
> Human-Computer Interaction

"Du et al. (2014) demonstrated 21 distinct compound expressions from non-linear AU combinations"

The 4 Challenges

AU Complexity
Non-linear Action Unit combinations cannot be decomposed additively

Semantic Ambiguity
"Sadly fearful" vs. "sadly angry" look nearly identical visually

Contextual Dependency
Requires semantic understanding beyond low-level pattern matching

Data Scarcity
RAF-CE: only 3,618 images, 14:1 class imbalance

Traditional CNN-based FER achieves only 42.3% on compound emotions

Research Gap & Our 5 Contributions

No prior work has systematically applied vision-language models to compound FER

Comparison :

Prior best (Ma et al.)	63.2% on AffectNet
Our result	96.81% on RAF-CE ✓ VLM ✓ Explainability

Contributions :

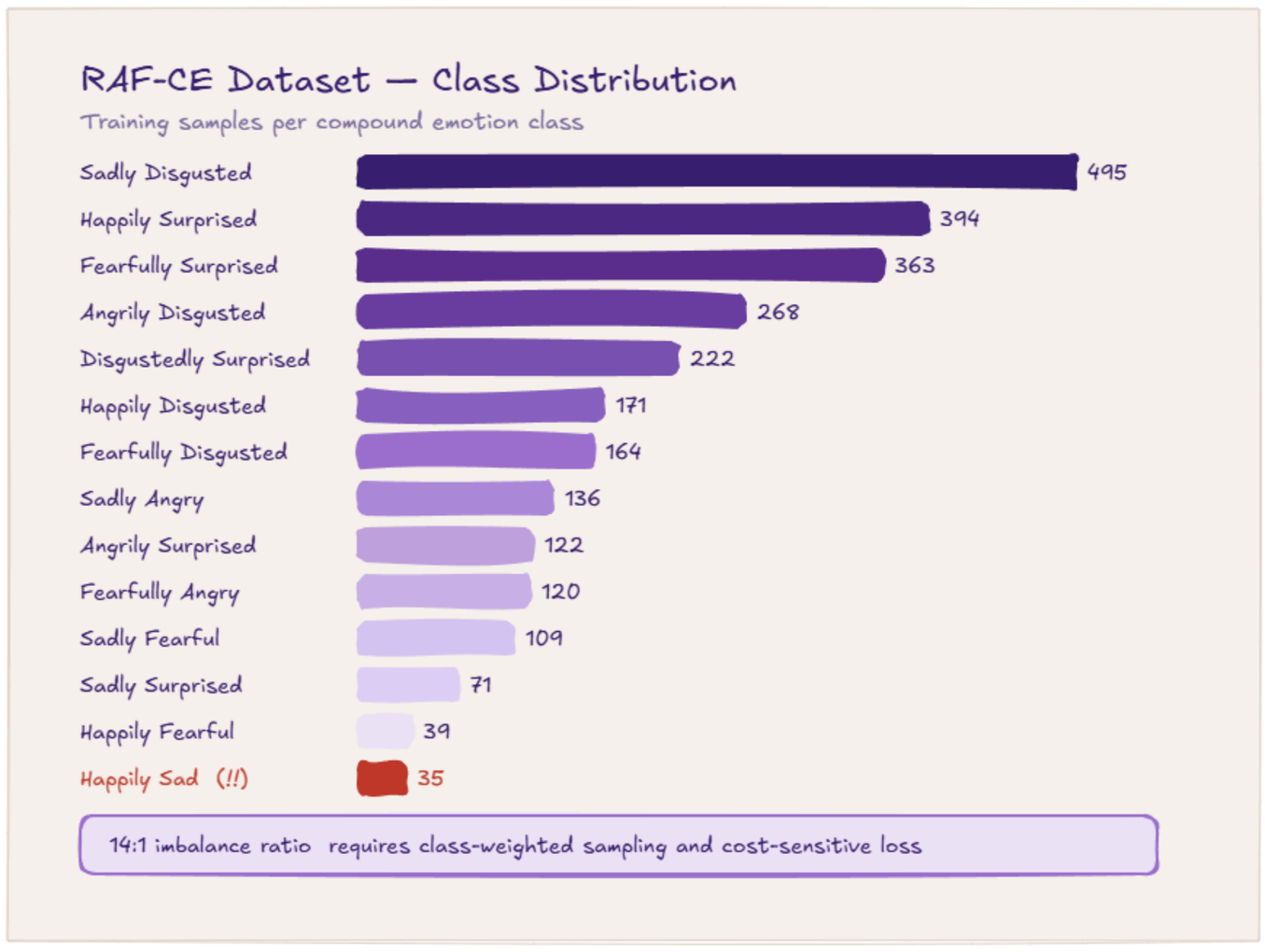
- First **multi-architecture** benchmark: CNN → Transformer → VLM on RAF-CE
- Novel VLM adaptation: frozen extraction (**LLaVA**) + LoRA fine-tuning (**BLIP-2)**
- **96.81**% accuracy — **117**% improvement over **CNN baseline**
- Comprehensive **Grad-CAM** & **SHAP** explainability
- Production-ready **Streamlit** web application

RAF-CE: Real-world Affective Faces with Compound Emotions

RAF-CE Dataset — Key Statistics

3,618	14	909	14:1
TOTAL IMAGES	COMPOUND CATEGORIES	TEST SAMPLES	CLASS IMBALANCE

14:1 imbalance ratio → requires class-weighted sampling & cost-sensitive loss



Preprocessing & Class Balancing

1. **MTCNN Face Alignment**
   Detect & align
2. **Resolution Normalization**
   224 / 336 / 384 px
3. **Data Augmentation**
   Flip, rot, jitter
4. **ImageNet Normalization**
   mean 0.485...

WeightedRandomSampler

$w_i = 1 / N_c$ per sample

Balanced batches | Equal class freq.

Oversamples minority classes at batch level
no new data generated

Cost-Sensitive Cross-Entropy

$L = -(1/N) \sum w_{y_i} \log p_{y_i}$

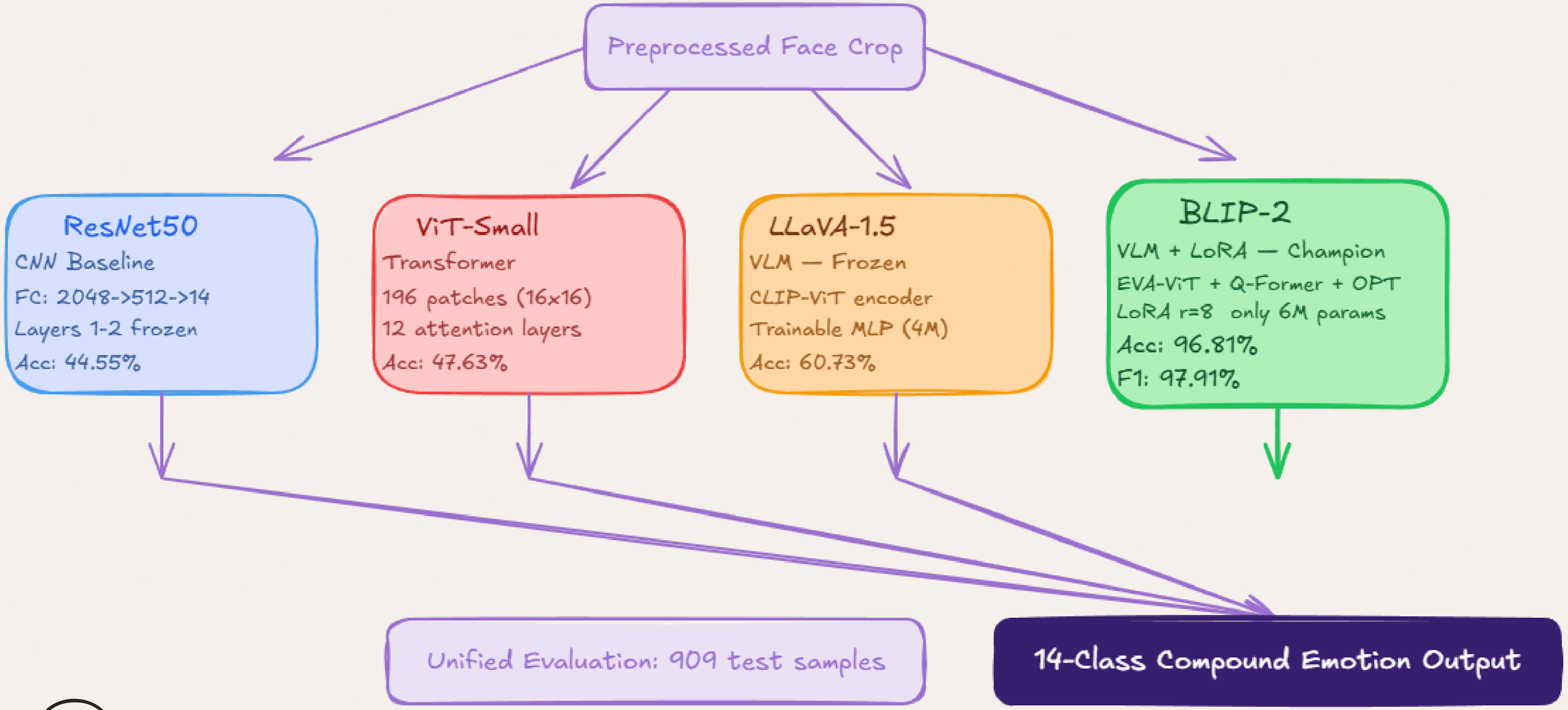
Rare class penalty | Weight by scarcity

Penalizes minority misclassification harder
during backpropagation

Combined: tackle 14:1 class imbalance — sampling + loss function

Multi-Architecture Pipeline Overview

Same input, 4 architectures, 14-class compound emotion output



From pure vision → to vision + language: how does adding semantic understanding change performance?

BLIP-2 wins: +117% vs CNN baseline

BLIP-2: The Champion Architecture

BLIP-2 — Champion Architecture

Face Image 384x384

↓

EVA-ViT-g/14 (FROZEN — 1B params)

576 visual tokens

↓

Q-Former: 32 Learnable Queries (LoRA r=8, a=16)

Cross-attention bridge → 32 output queries

↓

OPT-2.7B Language Model (LoRA + NF4 4-bit)

Constrained text generation

↓

Multi-Level Label Parser

↓

14-class Compound Prediction

Key Stats

Only 6M / 2.7B params trained

32 queries bridge vision to language

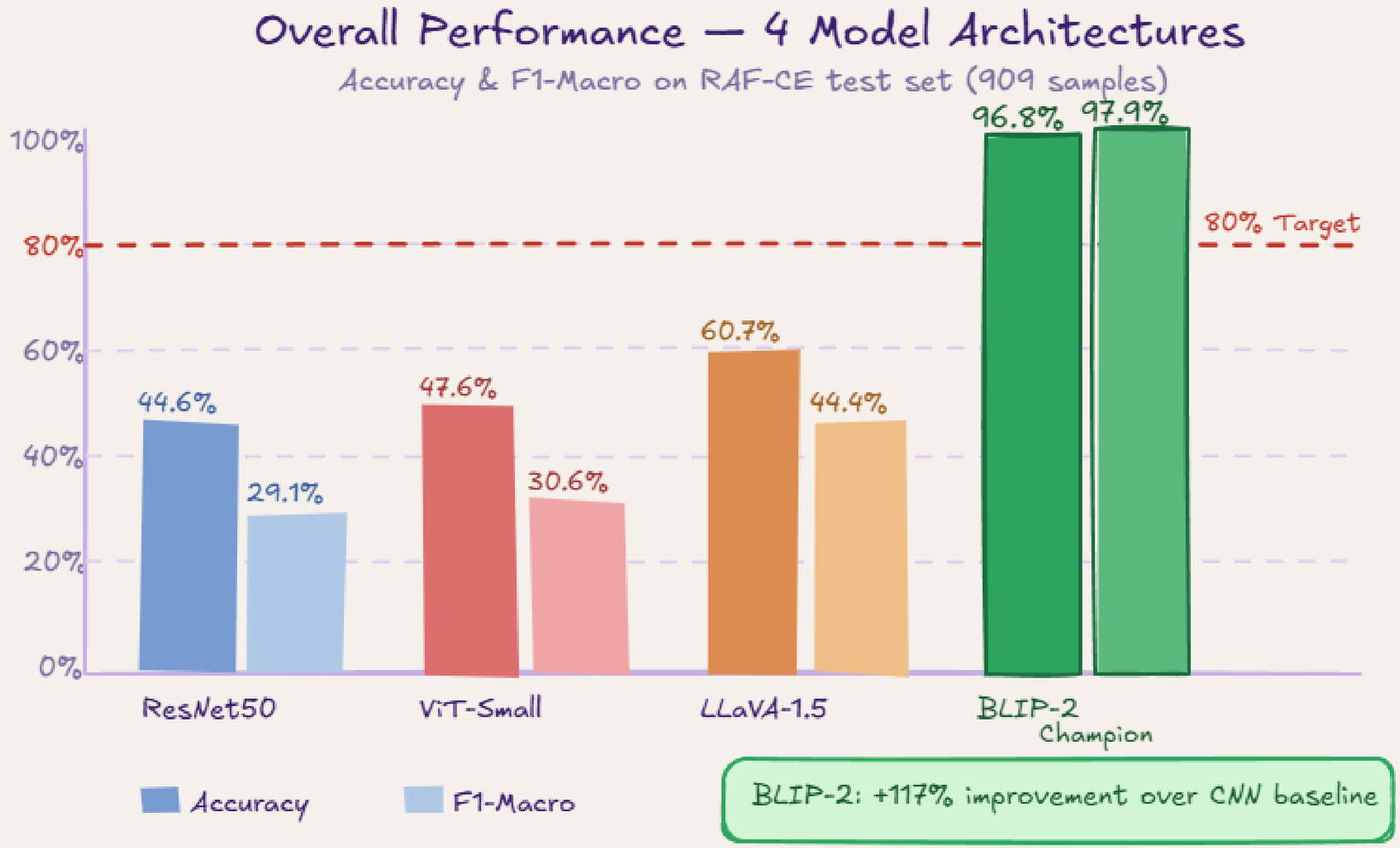
4-bit NF4 quant for efficiency

Pre-trained: 129M

Parser Tiers

Exact match — 90%
Both words — 70%
Single word — 50%
Fallback — 30%

Accuracy: 96.81%



Model	Accuracy	F1	Precision	Recall
ResNet50	44.55%	29.09%	30.94%	28.57%
ViT-Small	47.63%	30.60%	31.40%	30.35%
LLaVA-1.5	60.73%	44.40%	45.18%	45.22%
BLIP-2	96.81%	97.91%	97.65%	98.54%

Failure Analysis

Why Happily Disgusted = 0.84?

Conflicting semantics:
happy vs. disgusted create ambiguity in OPT embedding

Precision drops to 73%
but Recall = 100%
catches all cases correctly

Key Insight
Minority classes with only 6 test samples still reach F1 = 1.00 (Fearfully Disgusted)
Semantic grounding beats data scarcity

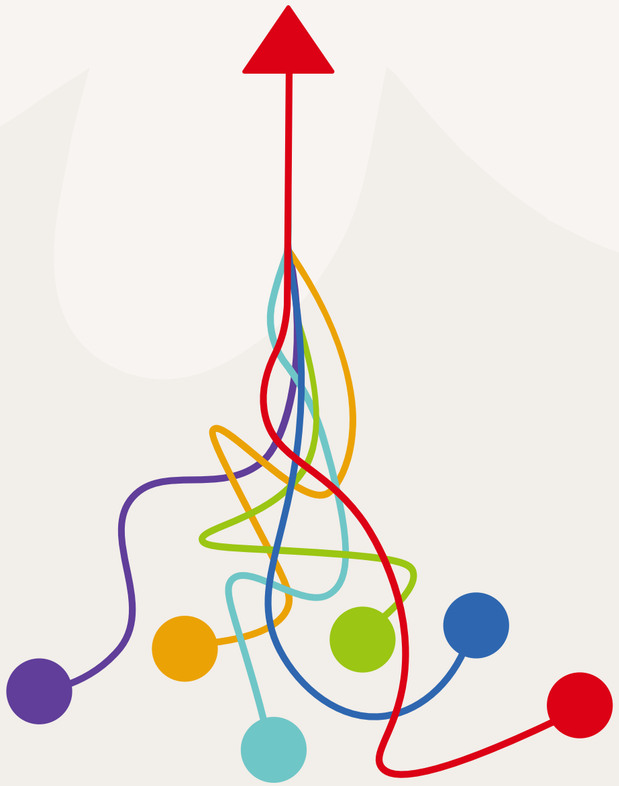
Legend
- Perfect F1 = 1.00
- Strong F1 >= 0.97
- Weak F1 = 0.84

9 of 14 classes = perfect score

Per-Class Performance & Failure Analysis

Training Convergence — Val. Accuracy vs Epochs

BLIP-2 converges fast — pre-trained on 129M image-text pairs

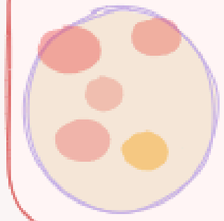


Explainability: Grad-CAM & SHAP

Grad-CAM Attention Maps

ResNet50

Scattered



Attention scattered across entire face — no specific AU focus. Background regions activated.

No anatomical grounding

ViT-Small

Diffuse global

Broad diffuse attention — whole face lit up equally. Some AU regions visible but imprecise.

Low spatial precision

BLIP-2

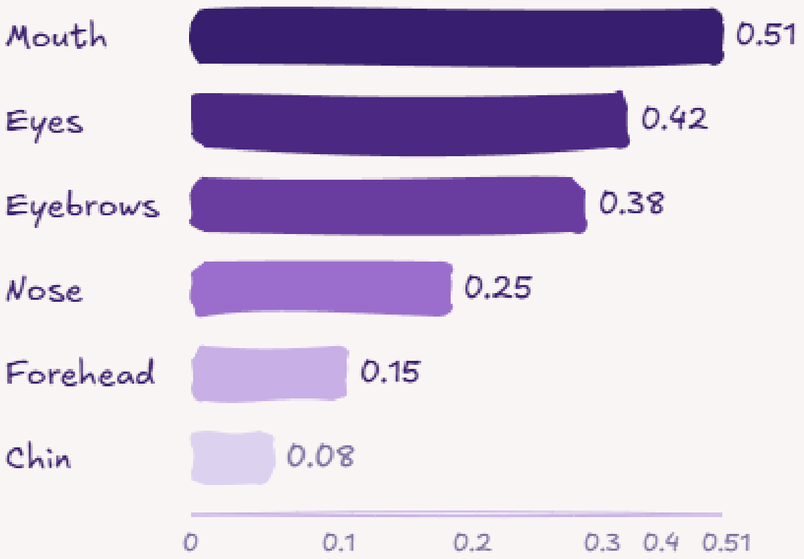
AU-precise

Precise AU1/2 activation: eyebrows highlighted. AU6/12: mouth corners perfectly localized.

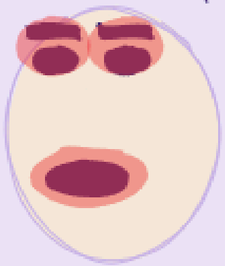
Matches FACS anatomy

SHAP Feature Importance

Contribution of each facial region to BLIP-2 prediction



Attention Map



Clinically Trustworthy

Mouth + Eyes drive all predictions

Low Chin & Forehead = no background shortcuts learned

Aligns with FACS AUs

Mouth & Eyes = primary discriminators — consistent with FACS. Low Chin/Forehead = no background shortcuts

Why VLMs Win + Limitations

1 Semantic Grounding

The knowledge foundation

Emotion concepts encoded deep in the language-vision joint embedding space before any fine-tuning occurs.

Rich prior knowledge

2 Q-Former Bridge

The smart filter

Cross-attention distills only the most emotion-relevant visual features into a language-aligned representation.

Noise suppression

3 LoRA Efficiency

Surgical parameter update

Rank-8 adapters injected into attention layers. Pre-trained knowledge stays perfectly intact. No forgetting.

Memory efficient

4 Multimodal Reasoning

Language as disambiguator

Language embedding space separates visually identical compound emotions that pure vision models cannot split.

Semantic disambiguation

Limitations

F1=0.84 on Happily Disgusted

Conflicting semantics

Tesla T4 hardware

Limited batch sizes

LLaVA token mismatch

Fallback to features — 60.7%

Conclusion & Future Directions

96.81%
BLIP-2 Accuracy
Champion result

97.91%
Macro F1 — all 14 classes
Across all categories

+117%
Gain over CNN baseline
44.55% -> 96.81%

VLMs are the dominant paradigm for fine-grained compound emotion recognition

Future Directions

1. **GPT-4V zero-shot evaluation**
   Can VLMs generalise without fine-tuning?
2. **Temporal compound FER from video**
   Emotion transitions captured over time
3. **Cross-cultural AU validation**
   Generalisation across ethnicities
4. **Ensemble BLIP-2 + AU-detection**
   Hybrid architecture for robustness

Live Deployment

APP

- Streamlit web application
- Real-time face inference
- 4-bit NF4 quantization
- Runs on consumer GPU

Production-ready today

Thank You

Any Questions?

Let's shape a smarter world — together.